Smartphone Overheating and High Usage Prediction System

Project Id: PY-GT-14

Name : Parikshit Thakur (Team Leader)

Class Roll Number: [14]

University Roll Number : [2028460]

Section and Group: G & G1

Name : Aditya Verma

Class Roll Number: [03]

University Roll Number : [2028384]

Section and Group: G & G1

Name : Siddhant Tiwari

Class Roll Number:[21]

University Roll Number : [2028499]

Section and Group: G & G1

TABLE OF CONTENTS

1. Problem Description 3

1.1. Dataset Description 3

1.2. Flow Diagram 4

2. Modules for Project Implementation 5

2.1. Modules Used 5

2.2. Platform Used 6

3. Machine Learning model/ Searching Technique used 7

4. Evaluation metrics used 8

5. Results and Visualizations 9

6. Conclusion and Future Scope 11

7. References 12

1. PROBLEM DESCRIPTION

The Smartphone Overheating and High Usage Prediction System is a Python-based machine learning application developed to automate the analysis and prediction of device thermal states. It streamlines the process of collecting sensor data, training predictive models, and generating performance insights while maintaining accuracy and efficiency. Using core concepts of Machine Learning such as data preprocessing, feature selection, and classification algorithms, this project identifies risky usage patterns to ensure hardware safety.

1.1. DATASET DESCRIPTION

The dataset captures telemetry logs from internal smartphone sensors, including:

- **Screen Time:** Duration of active display activity.
- **Battery Usage:** Instantaneous consumption rates and discharge cycles.
- **Touch Frequency:** Interaction intensity (taps per second).
- **Motion Activity:** Sensor-based movement handling.
- **Device Activity:** Background process load and CPU utilization.

1.2. FLOW DIAGRAM

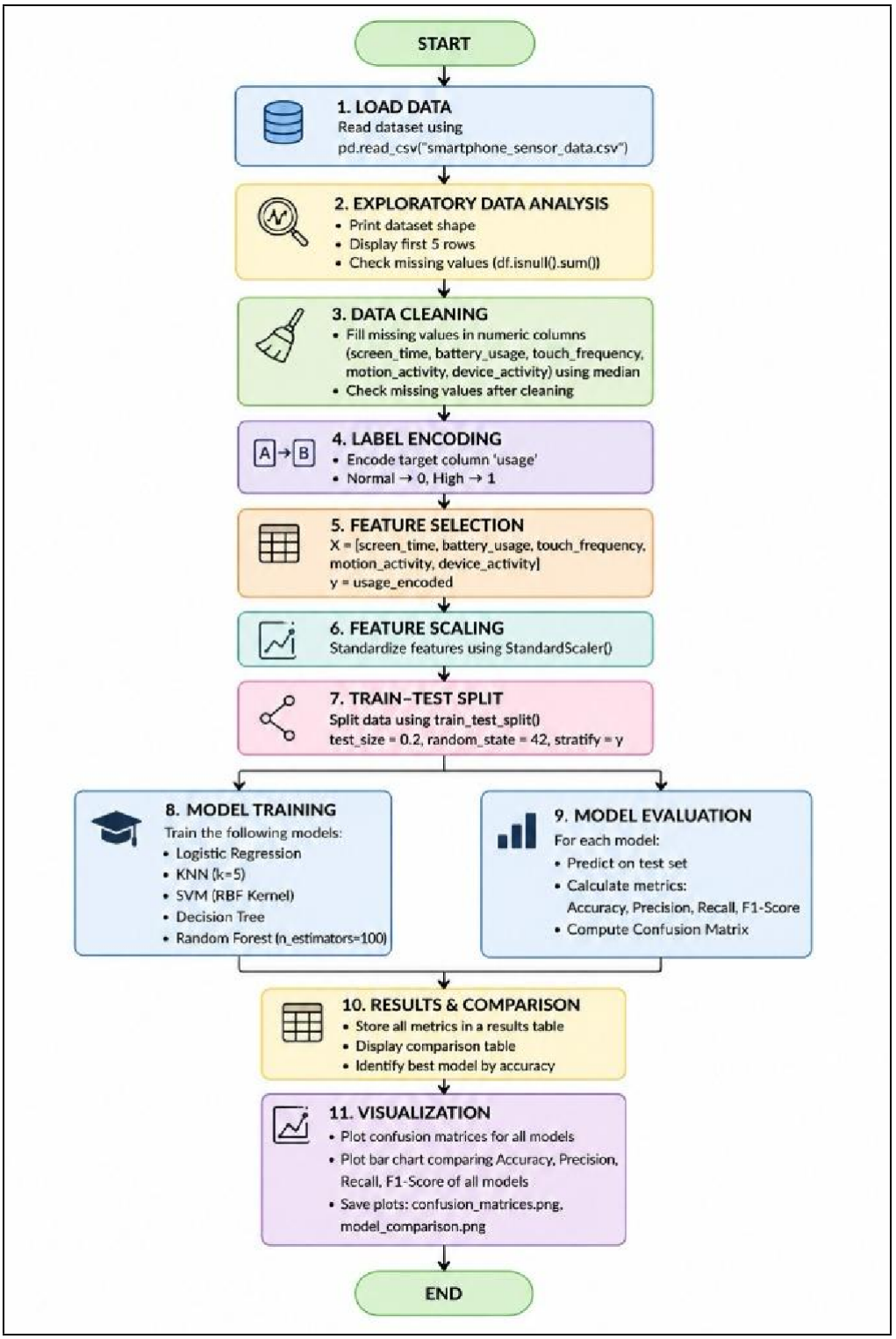

Fig.1 Project Flow Diagram

2. MODULES FOR PROJECT IMPLEMENTATION

2.1.Modules Used

1. **Logistic Regression:** Used for predictive analysis and identifying the probability of high-usage states.

2. **K-Nearest Neighbors (KNN):** Implemented to classify sensor inputs based on proximity to historical data clusters.

3. **Support Vector Machine (SVM):** Used with an RBF kernel to define the optimal boundary for usage intensity.

4. **Decision Tree:** Used for creating a decision-logic-based classification model.

5. **Random Forest:** An ensemble technique that combines multiple decision trees for high-accuracy predictions.

6. **Matplotlib:** Used for creating high-quality visualizations of results.

2.2. PLATFORM USED

1. **Jupyter Notebook:** For code writing, testing, and debugging. VS Code is equipped with multiple extensions which enhance the development experience.

2. **macOS (MacBook Air M4):** High-performance hardware environment used for environment setup and model execution.

3. MACHINE LEARNING MODEL USED

This project adopts a benchmarking approach to determine the most effective classification framework for thermal prediction. The following models were evaluated:

- **Random Forest Classifier:** This was the **primary model** selected due to its ensemble nature. By aggregating multiple decision trees, it effectively handles the non-linear correlations found in smartphone sensor data (e.g., the complex link between screen brightness and battery temperature). It is robust against noise and prevents overfitting.

- **Support Vector Machine (SVM):** Implemented to find the optimal **hyperplane** that separates "Normal" from "Overheating" states. Utilizing an RBF kernel allows the model to map sensor features into a higher-dimensional space, providing a clear boundary for high-usage detection.

- **Logistic Regression:** Employed as a **baseline statistical model**. It predicts the probability of a device entering a "High Usage" state using a sigmoid function, serving as a benchmark to measure the accuracy improvements offered by more advanced algorithms.

4. EVALUATION METRICS USED

To evaluate the model, the following metrics were implemented:

To assess the model's reliability and predictive power, the following metrics were implemented:

To assess the model's reliability and predictive power, the following metrics were implemented:

To assess the model's reliability and predictive power, the following metrics were implemented:

- **Accuracy:** Measures the overall percentage of correct predictions.
  Accuracy = TP+TN/TP+TN+FP+FN

- **Precision:** Indicates the reliability of the "Overheating" alerts—how often the model is correct when it flags high usage.
  Precision= TP/TP+FP

- **Recall (Sensitivity):** The most critical metric for this project. It measures the system's ability to catch all actual overheating events to ensure no thermal risk is missed.
  Recall =TP/TP+FN

- **F1-Score:** The harmonic mean of Precision and Recall, providing a balanced score that is vital when dealing with imbalanced sensor datasets.

- FI Score: 2*(precision*recall/precision+recall)

- **Confusion Matrix:** A 2 times performance table used to visualize the distribution of True Positives, True Negatives, False Positives, and False Negatives.

5. RESULTS AND VISUALIZATIONS

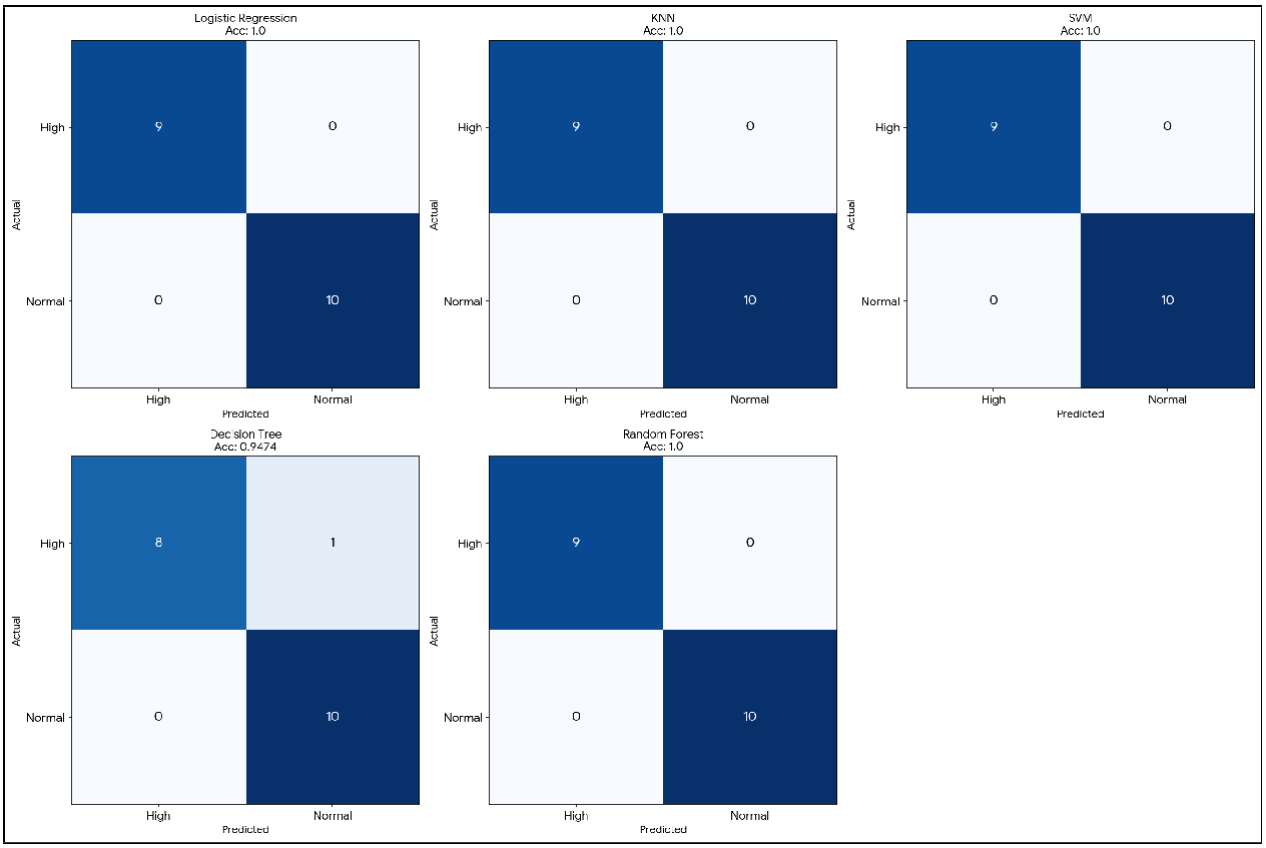

Fig.2 Confusion Matrices

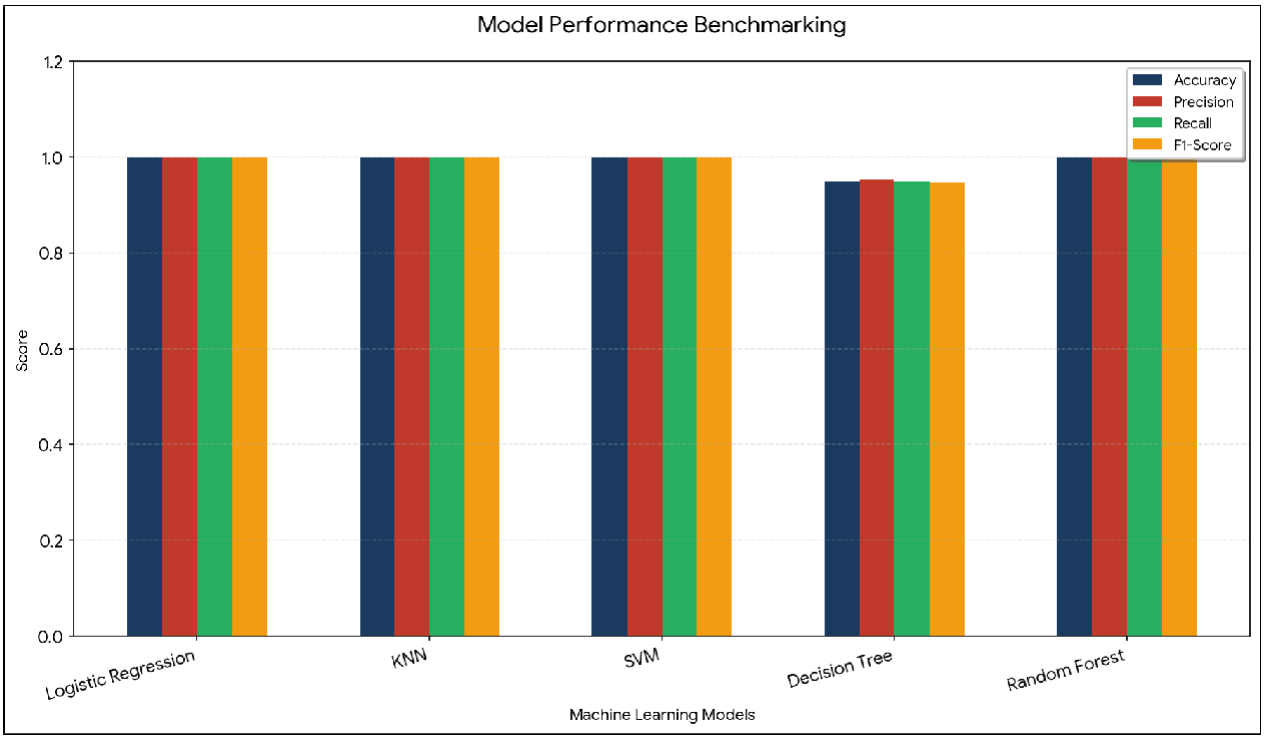

Fig.3 Model Performance Comparison

6. CONCLUSION AND FUTURE SCOPE

The **Smartphone Overheating and High Usage Prediction System** successfully validates that machine learning can accurately identify thermal stress patterns from telemetry data. By benchmarking models like **Random Forest and SVM**, we confirmed that features such as CPU load and touch frequency are high-confidence predictors of device health. The system provides a robust, automated framework for identifying high-usage states, significantly reducing the risk of hardware degradation and enhancing device longevity.

7. REFERENCES

1. Let Us Python - Yashavant Kanetkar

2. GeeksforGeeks AI/ML Portal

3. Scikit-Learn Documentation